ECON 100B: Macroeconomics

Class 13: Inflation (Jones Chapter 8)

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Agenda

- Announcements
- Current events
- Wrap up Jones Chapter 7: The Labor Market
- Jones Chapter 8: Inflation

Announcements

- Take-home Midterm Essay released 3/8, due 3/20
    - Scores will be high
    - You can use many tools including AI, but essays written solely by AI typically lose points and break the rules. Up to 10% of ECON 157 for example. AI is not a student in this classroom and won't know the best answers

- Spring Break 3/23-3/27

Current events

- Inflation

Inflation measures

- Consumer Price Index (CPI)
  - Most commonly referenced
  - Collected with surveys of consumers conducted by the Bureau of Labor Statistics (the “unemployment folks”)
  - Omits some things purchased on behalf of consumers, notably health insurance premiums, thus bigger weight on shelter

- Personal Consumption Expenditures (PCE) deflator
  - Preferred by the Federal Reserve
  - Calculated by the Bureau of Economic Analysis (the “GDP folks”)
  - Measures the prices of ALL of “C” in $Y = C + I + G + NX$
  - (Other differences like updated methodologies and backcasting)

“Core” inflation

- Core CPI or PCE deflator just drop the components that historically have wiggled around a ton
    - Food prices, food is internationally traded and hit by weather
    - Energy prices, seriously you guys

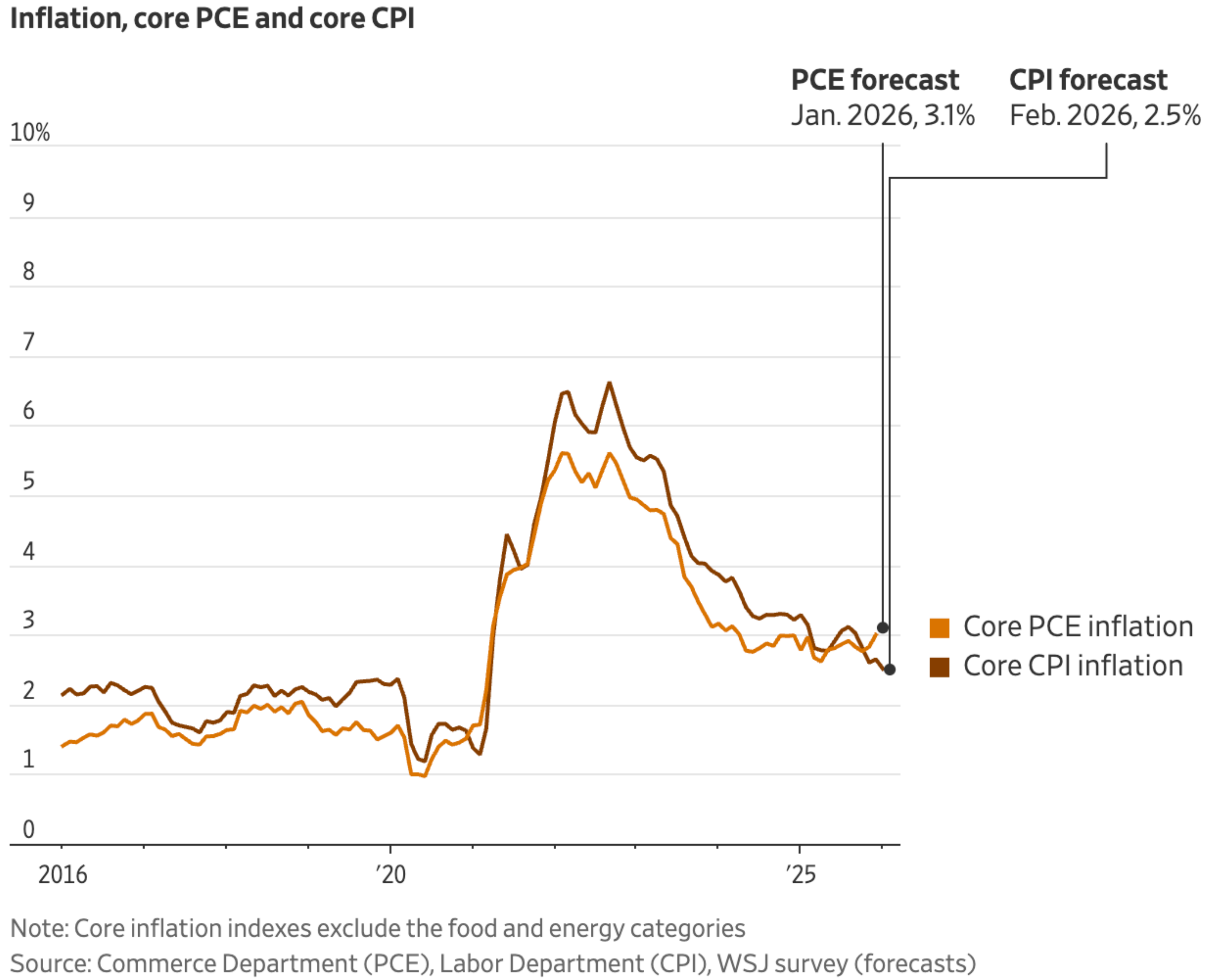
Inverted divergence between CPI and PCE

- Grossman (3/9/26) "Is Inflation Cooling or Stubbornly High? Both Can Be True." WSJ

- Prices of shelter are weighted higher in core CPI and have been "cooling"

- Health insurance premiums have a bigger weight in PCE deflator and have been "heating up"

- Upends usual π-CPI > π-PCEd



The oily elephant in the room …

Wallace (3/8/26) "The Long-Feared Persian Gulf Oil Squeeze Is Upon Us," WSJ

… is not such a party pooper?

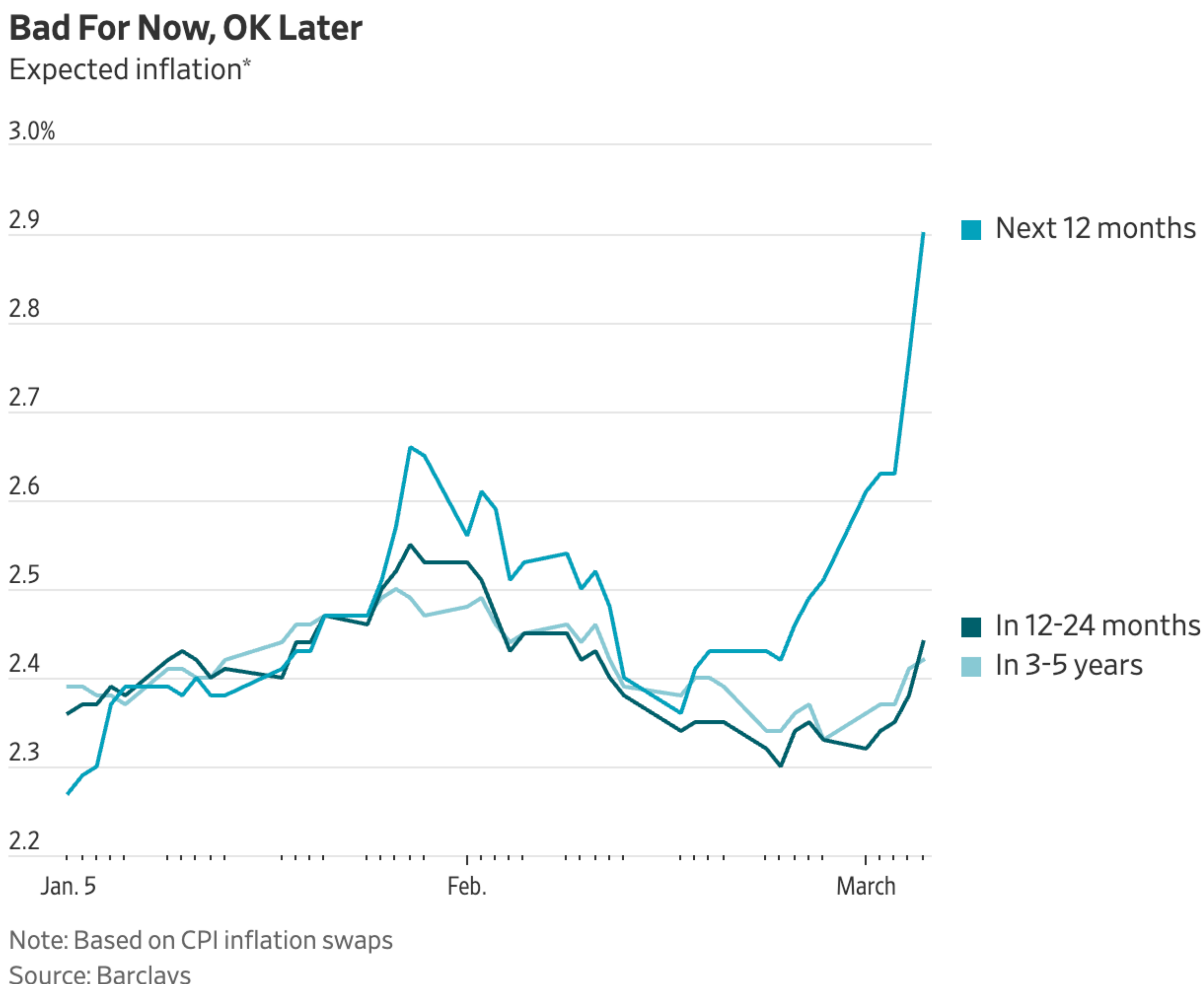
Ip (3/8/26) "Why the Oil Shock Probably Won't Derail the Economy. And One Way It Might." WSJ

- One way oil might derail the economy:

  If consumers stop believing inflation will fall

- The Fed has its work cut out for it

- (In the 1970s, it failed badly)



Learning objectives in Chapter 8 (1/2)

- Inflation π is the growth rate in the overall price level: $\pi = g[P]$

- The Fed operates differently now, but this basic view is useful historically and sometimes internationally:

$$M \cdot V = P \cdot Y$$

where

M is the stock of “money” or credit
V is the “velocity” of money use
P is the price level
Y is real GDP, so $P \cdot Y$ is nominal GDP

- If velocity is unchanging, then $\pi = g[M] - g[Y]$

- Our growth model gives us Y.
  Then inflation increases when money growth increases

- So governments should make sure M doesn’t increase too fast

Learning objectives in Chapter 8 (2/2)

- Nominal interest rates i are the cash rates of return over time, and they include a real interest rate R plus inflation π

$$i = R + \pi$$ called the Fisher equation

- This is the equilibrium nominal interest rate that will clear markets given levels of R and π. (*It's not a mechanism for controlling* π)

- Inflation is costly. Consumers hate it. Savers hate it. Borrowers might actually love it, because it erodes the real value of debts

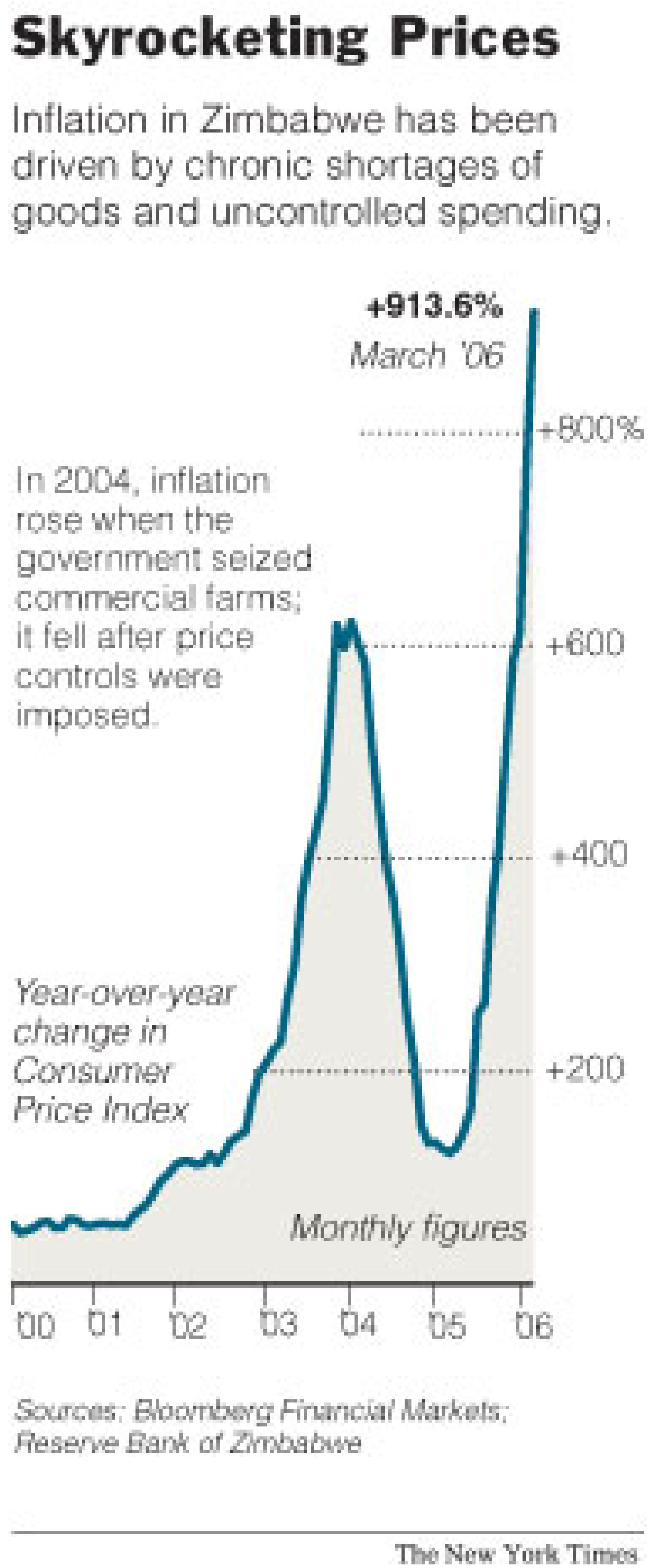
- Governments might print money to cover budget deficits, but that creates inflation and maybe hyperinflation

Inflation

New York Times, February 7, 2007:

"As Inflation Soars, Zimbabwe Economy Plunges."

- **Hyperinflation** reached an annual rate of 1,281 percent this month
- Seeking to revive farm production, the government sells gasoline to farmers at a bargain rate of 330 Zimbabwe dollars per liter — and farmers promptly resell it on the black market for 10 times that, **leaving their fields idle**.
- The **central bank's** latest response to these problems, announced this week, was to **declare inflation illegal**. From March 1 to June 30, anyone who raises prices or wages will be arrested and punished. Only a "firm social contract" to end corruption and restructure the economy will bring an end to the crisis, said the reserve bank governor.
- The speech was broadcast nationally. In downtown Harare, the last half was **blacked out by a power failure**.



The *New York Times*, May 2, 2006

How bad is inflation in Zimbabwe? Well, consider this: at a supermarket near the center of the capital, toilet paper costs $417.

No, not per roll. Four hundred seventeen Zimbabwean dollars is the value of a single two-ply sheet. A roll costs $145,750 — in American currency, about 69 cents.

The price of toilet paper, like everything else here, soars almost daily, spawning jokes about an impending better use for Zimbabwe's $500 bill, now the smallest in circulation.

Zimbabwe's inflation is hardly history's worst

In Weimar Germany in 1923, prices quadrupled each month, compared with doubling about once every three or four months in Zimbabwe

More recent events in Zimbabwe

- NY Times 11/2/2008:
  - Inflation in Zimbabwe was running around 230 million percent per year, according to the UN, which is about 5% *per day*
  - Zimbabwe's Reserve Bank suspended electronic payments and limited cash withdrawals; you had to *write checks*, and by the time the check clears, say 3 days, its value has dropped more than 15%
- From Wikipedia entry and Bloomberg news 4/20/09:
  - On 1/12/09, Zimbabwe introduced the $50,000,000,000 note
  - Later that month, plans for $100 trillion note
  - In part due to political reshuffling (power sharing between Mugabe and Tsvangirai, and a new economics minister), a decision to abandon the ZWE-$ and use South African rand and the U.S. dollar
  - Inflation was –3% in March, the second consecutive monthly decline

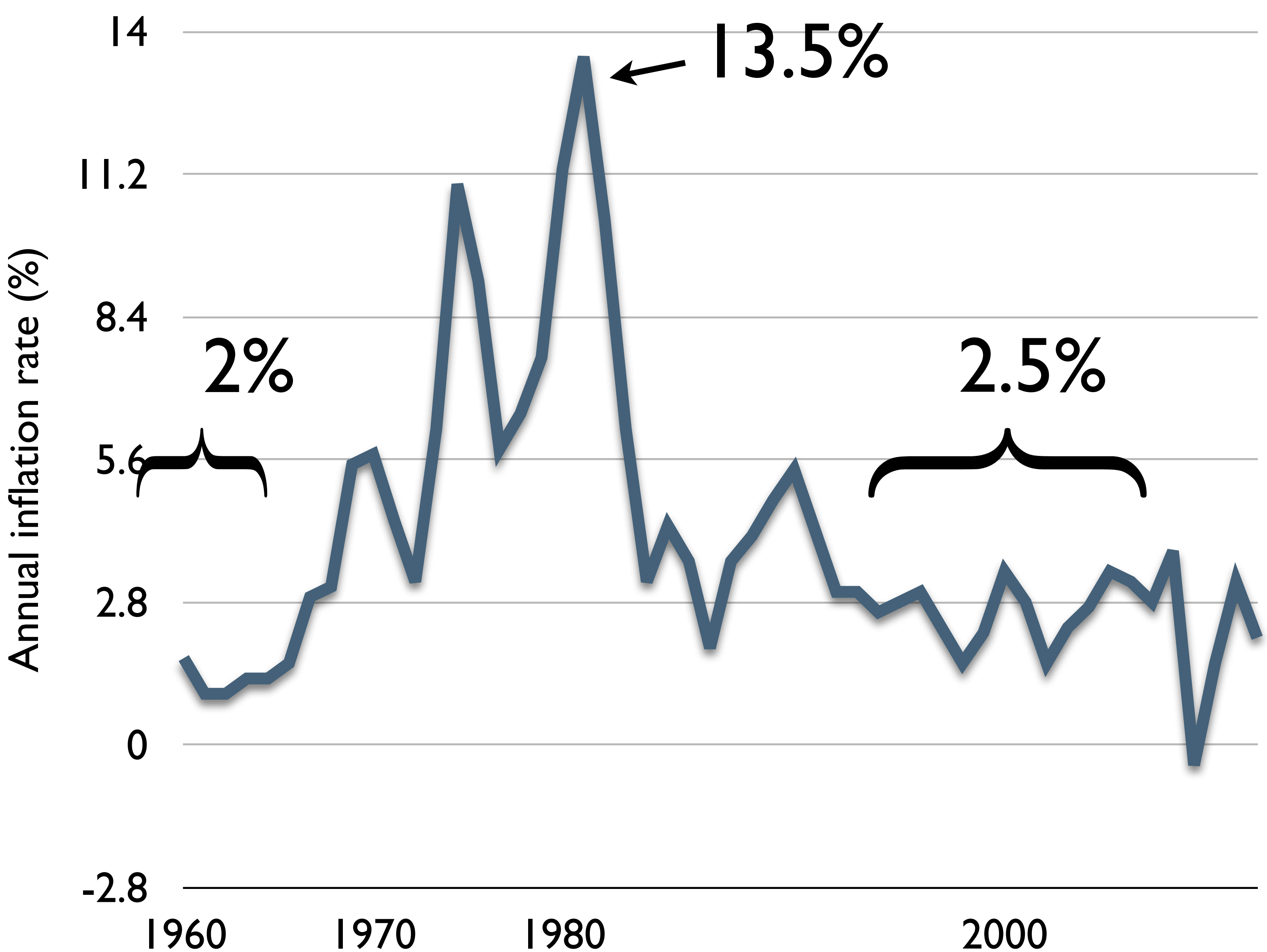
What is **inflation**?

- Inflation is the rate of increase in prices over time
- If P_t is the price level at time t, then inflation, π, is its growth rate:

$$\pi = g_P = \frac{\Delta P_t}{P_t}$$

- Inflation measures the rate at which the average price is rising
- If your income is not rising as fast as inflation, then your power to purchase the average good is deteriorating!
- How has inflation behaved in the U.S. over time?

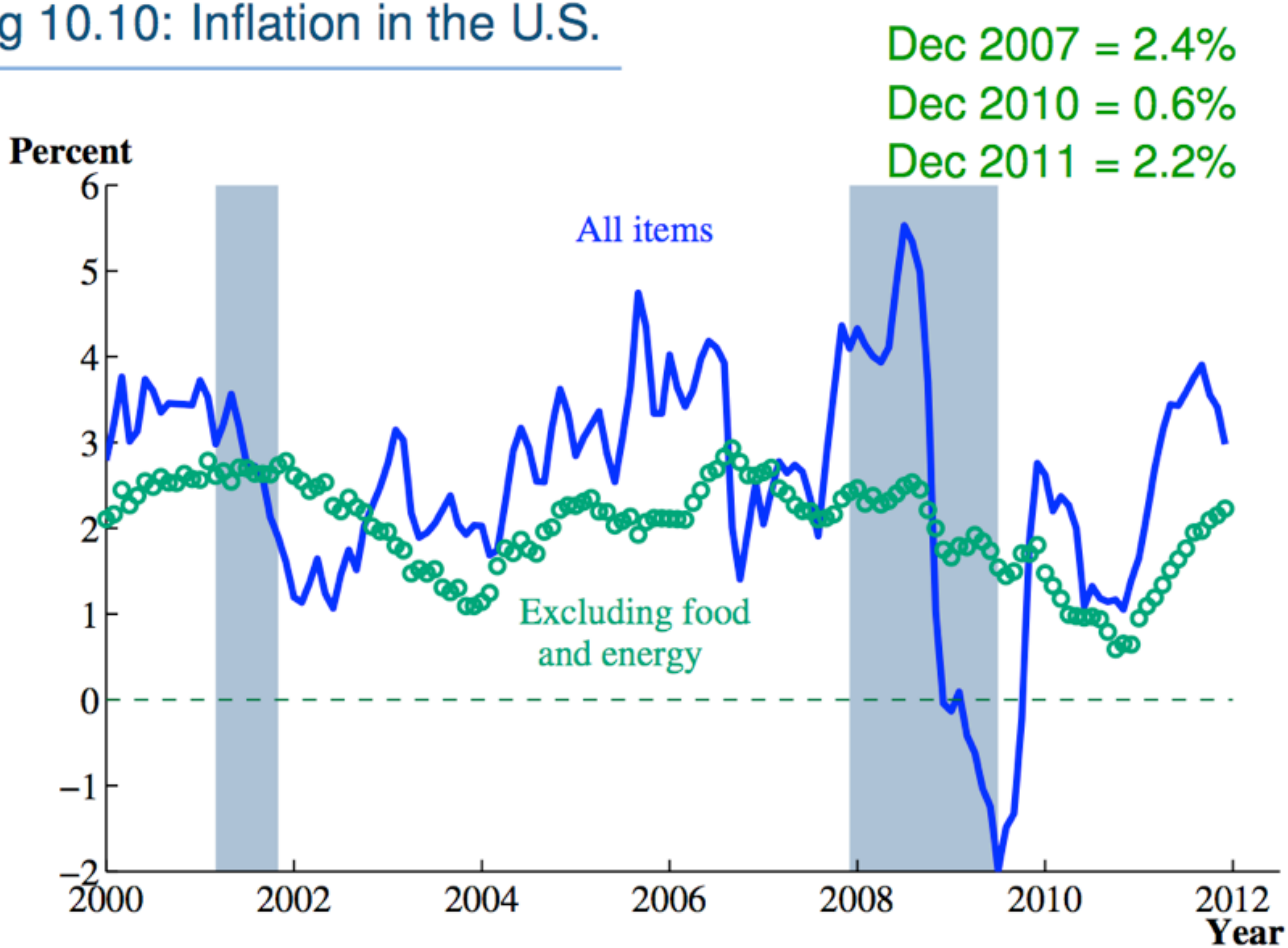
During recent history, inflation has been low, then **extremely high** and variable, and then low again



- In the 1960s, inflation was **low** and not particularly volatile, around 2% per year
- But as the Vietnam War escalated, and as the oil shocks hit the world economy in the 1970s, inflation **increased** and became considerably more volatile, peaking in 1980 at 13.5%
- Then it fell, and for the last 10-15 years, inflation had been relatively **subdued...**

Recently, inflation briefly became **deflation**! (negative inflation)

Fig 10.10: Inflation in the U.S.



Why did the "Great Inflation" in the late 1970s happen?

- Part of the reason involves the oil shocks — that also increased unemployment worldwide during this period
- But part of the reason, it is now widely agreed, involved **bad policymaking**
- How? <u>The central bank was "printing" too much money</u>
  - The central bank, **the Federal Reserve** in the U.S., regulates the money supply, while the U.S. Mint actually prints money
  - *Technically, the central bank doesn't really set the money supply, it sets a target nominal interest rate* — we'll talk about this in later chapters
- To understand when and why too much money actually becomes "too much," we need the **Quantity Theory of Money**

But first, *what is money*?

- The cash in your pocket is obviously money, right?

- In modern economies, the cash in your pocket is nothing but a credit, a medium of exchange, <u>with no instrinsic value</u>. Historically, money was legally linked to amounts of gold, but that is no longer the case

- What else is credit? *Bank accounts*, credit cards, IOUs...

- We'll define money — usually in terms of the <u>size of the **money supply**</u> — based on how *liquid* it is: how rapidly you can get it to trade for goods and services

- From most liquid to least, money consists of:

Cash, Banks' reserves, Checking accounts, Savings, money market accounts

How do we move from money to inflation?

- Money is a *nominal*, not a **real** quantity: your $10 bill will buy only so many goods as you can afford given their prices

$$\$10 = p \times q$$ = $2 per latté times 5 lattés

- This simple relationship is *almost* all we need to know in order to compare the **nation's** money supply to the **nation's** P and Q —

  which are the price level and real GDP, so PxQ = nominal GDP

- The <u>extra twist</u> is that in the economy, the same money gets used multiple times during a year

- We use the term "velocity" and the letter "V" to measure this

- So for the aggregate economy: $$M \times V = P \times Y$$

- The money supply (M) times velocity (V) is equal to the price level (P) times real GDP (Y), where V = usages of the $10 bill per year

What does this quantity theory of money tell us about the price level?

$$M_t V_t = P_t Y_t$$

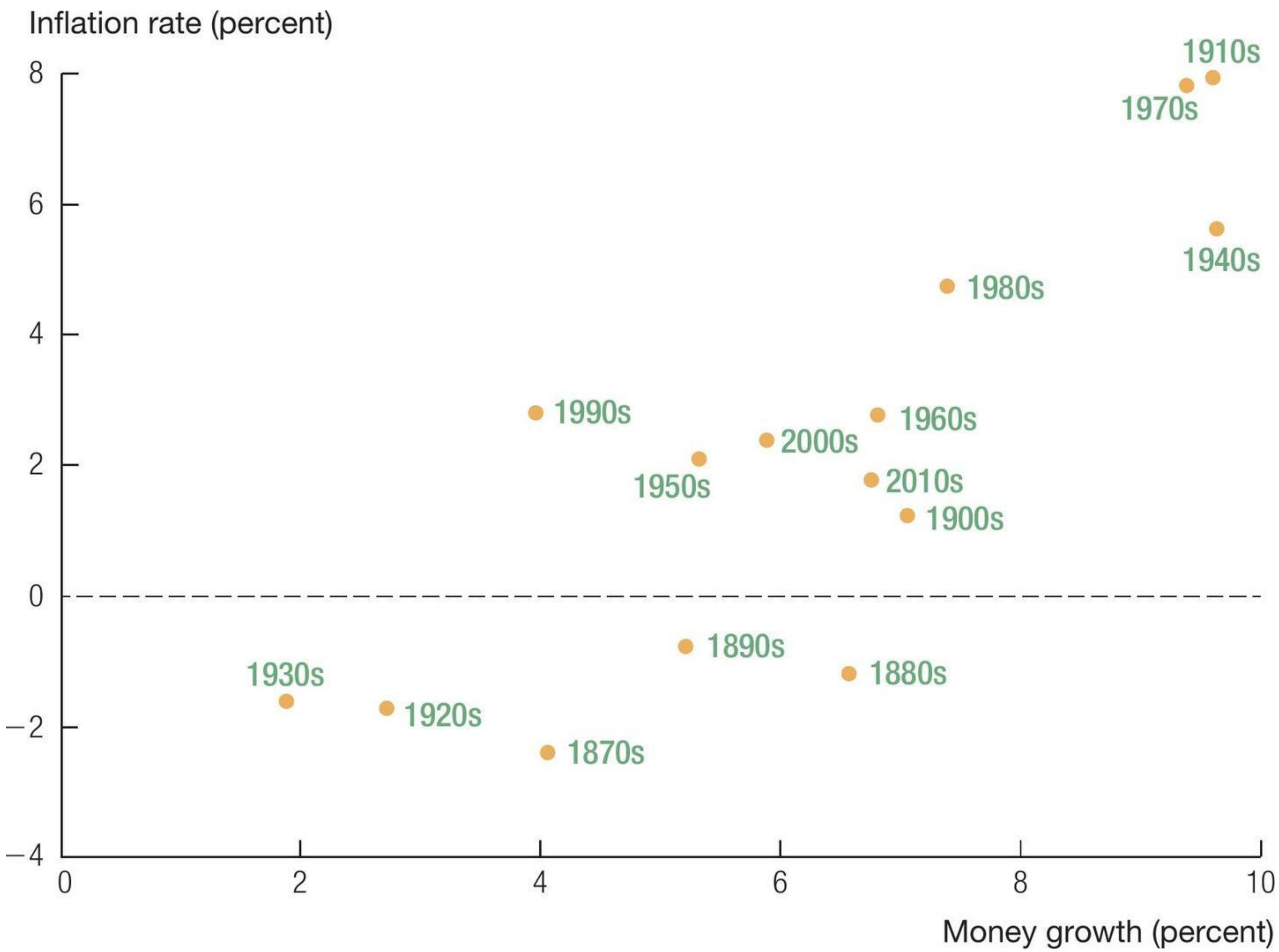
- Rearranging for P_t, we find:

$$P_t = \frac{M_t V_t}{Y_t}$$

- In the long run,
  - Velocity (V) is some constant (It could change in theory, but it tends to be basically constant)
  - Real GDP (Y) is growing steadily and fairly predictably, so at *t*, it is just some given level
- Under these conditions, changes in the money supply (M_t) will feed directly into changes in the price level (P_t)
- By extension, the central bank can manipulate *growth* in the money supply and thus affect *inflation*...

Take the growth rate of both sides of $P_t = \frac{M_t V_t}{Y_t}$ where $g_V = 0$

$$g_P = \pi = g_M - g_Y = \%\Delta M - \%\Delta Y$$



Sources: Money growth rates are computed using the data on M2 from the *Historical Statistics of the United States* (Cambridge: Cambridge University Press, 2006) and FRED. Inflation data are taken from measuringworth.com.

- The growth rate of real GDP, g_Y, is basically constant at 3.5% or so — just like we saw in the first 6 chapters
- What this equation says is that inflation is a function of the growth rate of the money supply
- In U.S. data the relationship isn't exactly a straight line
- Other countries show similar trends

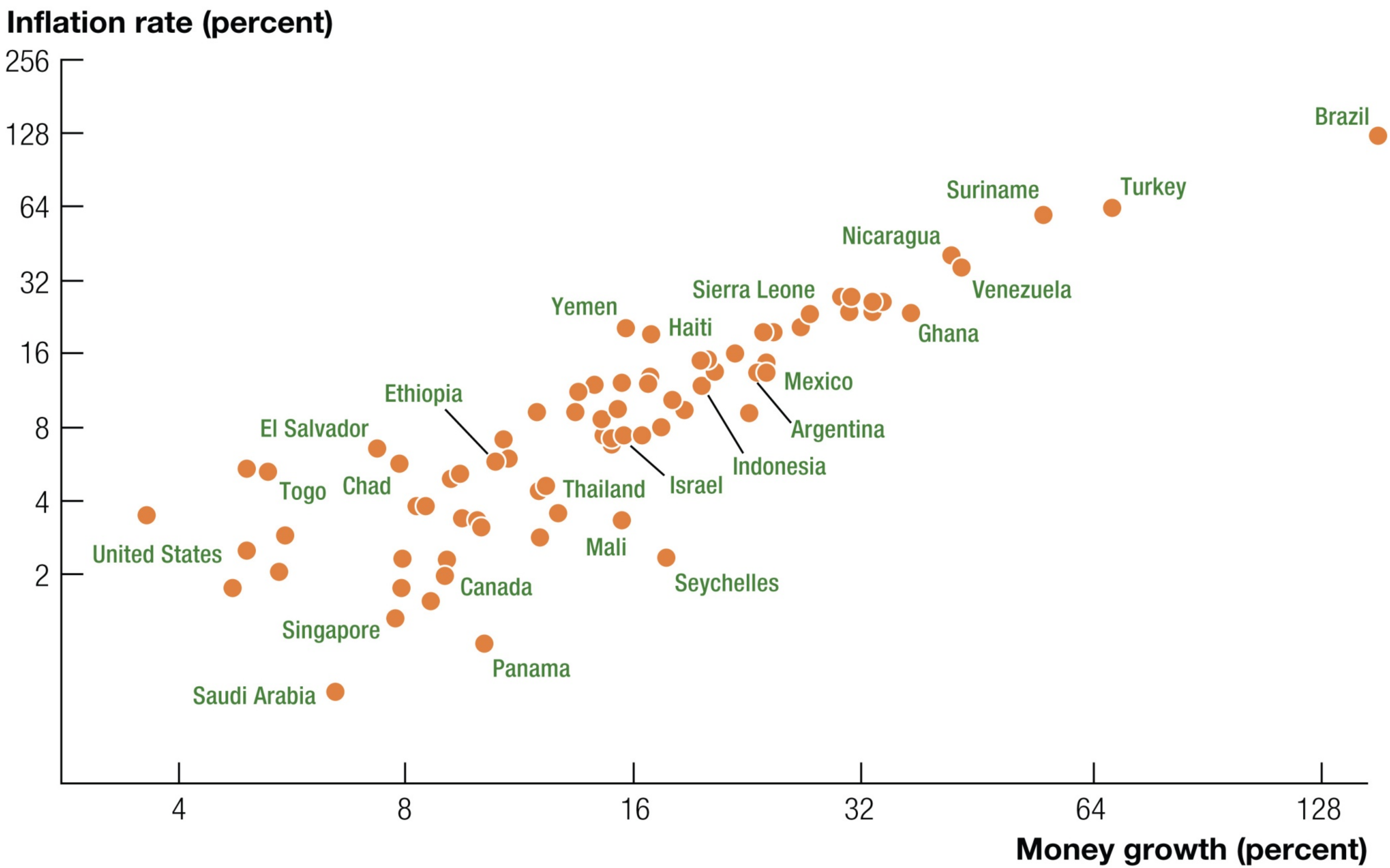
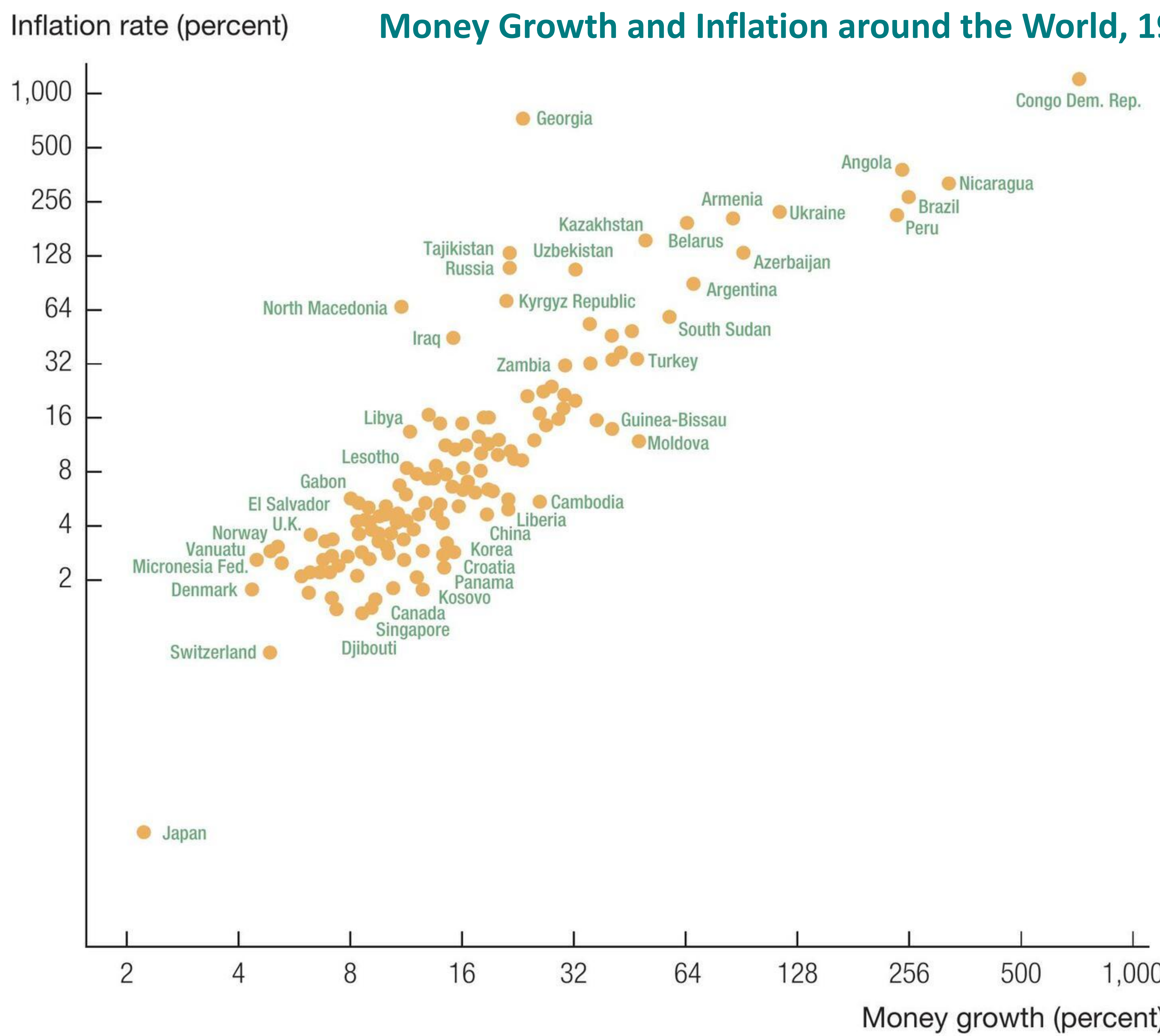


FIGURE 8.3 Money Growth and Inflation around the World, 1990–2003

Money Growth and Inflation around the World, 1990–2021



Source: World Bank, *World Development Indicators*. The figure plots the average annual percentage change in the price level (inflation) and money supply for a sample of countries. Ratio scales are used for both axes.

Interpreting Quantity Theory: MV = PY

- Think of money denominations and prices like the points associated with **scores in sports** like football or basketball

- What if each field goal in basketball were **4 points** instead of 2
  each free throw **2 points** instead of 1
  each three-point shot **6 points** instead of 3

  Would these changes have any effect on a team's wins and losses?

- No, it's hard to see any reason why, *if <u>all</u> points were scaled up by 2x*

- What's what here?

  - Points per score = price or "cash value" of score

  - Win or loss = real GDP

What does inflation affect?

- Changes in prices erode the real value of *any* nominal amounts
- **Labor income**:
  - Suppose you were making \$50,000 a year in 2006; you worked 40 hours a week x 50 weeks = 2,000 hours, and your wage was \$25 per hour
  - What if in 2007, the prices of everything you bought went up by 3% — but you were still making \$50,000 a year?
  - Your nominal income is the same, but your <u>real income has fallen</u> by 3%, and you could think of this as a decrease in your real wage rate of 3% (why? because of this next part:)
- **Capital income**:
  - Suppose you had bought a **corporate bond**: you paid \$100 to a company in return for a one-year bond which pays \$107 after a year
  - That's a 7% nominal interest rate. But if inflation is 3%, what's the real value of the bond payoff?

4%

Your bond that costs \$100 today pays you \$107 next year — a nominal interest rate of 7% — but inflation is 3%.

What is your real return on the bond?

One way we could proceed is to deflate \$107 by a growth factor of 3%

real return = 107 ÷ (1 + 0.03) = 103.9 = 3.9% real return

But when interest rates and inflation are *small*, we can use our growth rate rules and get *close enough* to the right answer:

growth in the real value of the bond =
growth in (the nominal value *divided by* the inflation factor)

the real interest rate = the nominal interest rate *minus* inflation

$$r = i - \pi$$

4% = 7% – 3%

Real and nominal interest rates have fluctuated over time, not always together, because inflation has fluctuated over time

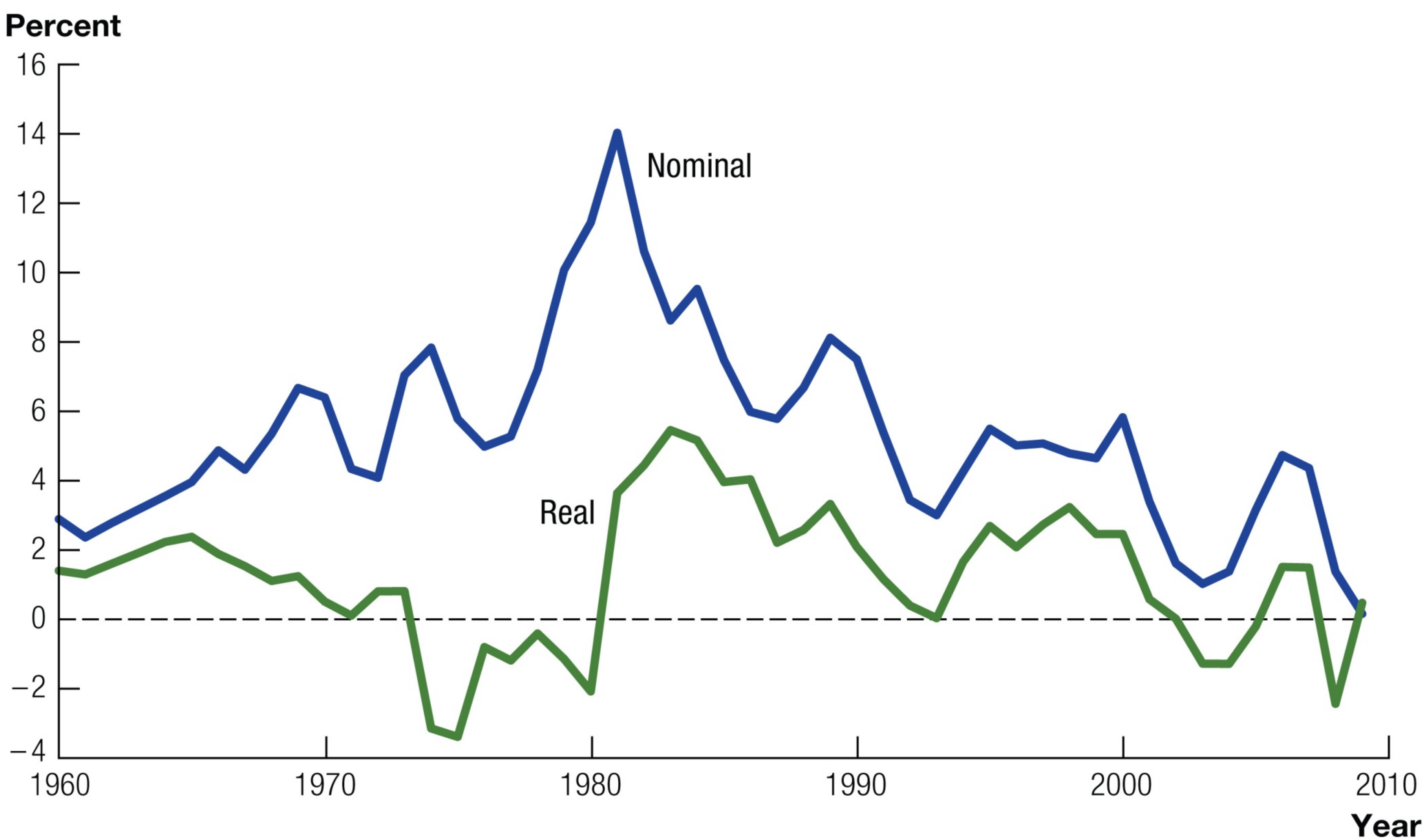


FIGURE 8.4 **Real and Nominal Interest Rates in the United States, 1960–2009**

- The gap between the nominal and real rates is the inflation rate: $r = i - \pi$
- We saw in our study of growth that the real interest rate is the marginal product of capital (MPK) — but we see here that the real interest rate has been ***negative!*** How could that be?
- Sometimes, the real interest rate departs from the MPK, and we'll see why soon

Inflation can fluctuate a lot — that's when it becomes costly, when we can't predict it

- Who is particularly exposed to unexpected inflation?
- Retirees with pensions that are fixed in nominal terms (Social Security is actually indexed to inflation, so that money is safe from inflation)
- Borrowers and lenders who agree to nominal interest rates — when inflation increases (decreases) unexpectedly, the real value of interest payments declines, which helps (hurts) borrowers
- Workers earning minimum wage — because it's a *nominal* wage target
- Upper middle-class families who may become hit by the Alternative Minimum Tax (AMT), which is not indexed to inflation
- Everyone who has to go to the bank more often because cash loses its value more rapidly ("shoe-leather costs")
- Everyone who has to keep updating their prices ("menu costs")

If inflation is costly, why would central banks (governments) ever print too much money?

- Unfortunately, printing money is a way to pay bills!
- Government can finance its spending (G) several ways:

$$G = T + \Delta B + \Delta M$$

through taxes (T), new bonds (ΔB), or new money (ΔM)

- The revenue from ΔM is called "seignorage" — "right of the lord or ruler." Back in the days of metal coins, kings stamped the metal into coins and kept some metal themselves!
- Who pays this "inflation tax?" People who hold currency
- (Printing money doesn't really work unless it is *unexpected*)

But in the 1970s, the Federal Reserve didn't want to collect seignorage, right? Why was inflation high?

- The best way to understand high inflation during the 1970s is by thinking about the quantity theory equations:

$$P_t = \frac{M_t V_t}{Y_t} \qquad g_P = \pi = g_M - g_Y = \%\Delta M - \%\Delta Y$$

- What if g_Y is smaller than you think — in other words, what if real output isn't growing as fast as you project?
- Inflation (π) will be higher than it otherwise would be!
- This is a pretty good explanation of the 1970s — productivity growth slowed quite a bit, and nobody knew it at the time — so money growth was too high, and so was inflation